Anna Sofia Moro, MSc Student in Physics

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Education

2023 – Present **M.Sc. Physics** Università degli Studi di Padova. *(Padua, Italy)*
Thesis title: *Characterization of Long Bent Silicon Crystals for Channeling Applications at the LHC.*
Supervisors: Prof. Marco Zanetti *(UniPd)*, Dr. Pascal D. Hermes *(CERN)*

2020 – 2023 **B.Sc. Physics** Università degli Studi di Padova. *(Padua, Italy)*
Thesis title: *Performance of the Final ENUBET Demonstrator Using CERN-PS Beams.*
Supervisor: Prof. Andrea Longhin *(UniPd)*

Master's Degree Path: High Energy Physics

Weighted Average: 28.875/30

Expected Score: 110/110

Completed Courses
- Theoretical Physics of the Fundamental Interactions
- Physics Laboratory
- Nuclear Physics
- Subnuclear Physics
- Introduction to Radiation Detectors
- Astroparticle Physics
- Advanced Physics Laboratory
- Experimental Subnuclear Physics
- Introduction to Research
- Machine Learning

Planned Courses
- Medical Physics

Projects & Research Activities

Jul. 2026 - Dec. 2026 **Short Term Internship - CERN** (Geneva, Switzerland).
Title: *Characterization of Long Bent Silicon Crystals for Channeling Applications at the LHC.*
Description: Evaluated the performance of bent silicon crystals for large-angle hadron deflection within the TWOCRYST project. Analyzed experimental beam test data to determine key channeling parameters, such as efficiency and deflection angle. Benchmarked simulation models against experimental results to assess crystal viability for future LHC fixed-target setups.

Mar. 2024 - Mar. 2025 **Advanced Physics Laboratory - Università degli Studi di Padova**.
Title: *Rutherford scattering measurements.*
Description: Quantified Rutherford scattering cross-sections across various deflection angles and target atomic numbers. Designed and implemented a Monte Carlo simulation to model the apparatus, optimize experimental parameters, and benchmark theoretical expectations against experimental data.

Projects & Research Activities (continued)

Jun. 2024 - Oct. 2024 — **Introduction to Research - Università degli Studi di Padova**.
Title: *Optimization studies for the measurement of* $\mathrm{BR}(\Lambda_b^0 \to \Lambda_c^+ D_s^{(*)-})$.
Description: Optimized the branching fraction measurement of the $\Lambda_b \to \Lambda_c^* D_s^-$ decay using LHCb data to constrain irreducible backgrounds in Lepton Flavour Universality tests. Employed advanced multivariate techniques for data analysis and conducted detailed evaluations of the associated systematic uncertainties.

Sep. 2023 - Dec. 2023 — **Bachelor's Thesis - Università degli Studi di Padova**.
Title: *Performance of the Final ENUBET Demonstrator Using CERN-PS Beams*.
Description: Analyzed data collected at the CERN-PS for the ENUBET experiment to develop a novel calibration method independent of reconstructed silicon tracks. Validated the performance of this new approach by evaluating the electron energy resolution of the demonstrator.

Skills

Technical

Programming	C/C++, Python, R
Physics Tools	ROOT, RooFit, Geant4
Data science	Machine Learning
Tools & OS	Git, LaTeX, Linux/Bash, MS Office

Language

Italian	Native speaker
English	Reading: C1, Writing: B2, Listening: C1, Speaking: C1
French	Reading: A2, Writing: A2, Listening: A2, Speaking: A2
German	Reading: A1, Writing: A1, Listening: A1, Speaking: A1